

Total member in the club (U) = 255

League of Legends (L) = 160

Super Smash Bros (S) = 133

Fortnite (F) = 112

$L \cap S = 80$

$S \cap F = 25$

$L \cap F = 56$

$L \cap S \cap F = 11$

1. To get the number of gamers that only play League of Legends (L), you take the total number of L players and subtract any players that play L as well as any of the other games. Taking away all players that play more than League of Legends from the total L players in the club will leave those that only play L:

$$\begin{aligned} L - (L \cap S) - (L \cap F) - (L \cap S \cap F) \\ = 160 - 80 - 56 - 11 \\ = 13 \end{aligned}$$

2. To get the number of gamers that only play Super Smash Bros (S), you take the total number of S players and subtract any players that play S as well as any of the other games. Taking away all players that play more than Super Smash Bros from the total S players in the club will leave those that only play S:

$$\begin{aligned} S - (L \cap S) - (S \cap F) - (L \cap S \cap F) \\ = 133 - 80 - 25 - 11 \\ = 17 \end{aligned}$$

3. To get the number of gamers that only play Fortnite (F), you take the total number of F players and subtract any players that play F as well as any of the other games. Taking away all players that play more than Fortnite from the total F players in the club will leave those that only play F:

$$\begin{aligned} F - (S \cap F) - (L \cap F) - (L \cap S \cap F) \\ = 112 - 25 - 56 - 11 \\ = 20 \end{aligned}$$

4. To get the number of players that do not play any of these three games, take the total number of gamers in the club and subtract all gamers that only play one of the three games and those that play any mixture of them. The total number of gamers in the club (U) includes those that do not play League of Legends, Super Smash Bros, or Fortnite so taking away all those that do play any one of them or any mixture of them would leave those that do not play any of them:

$$\begin{aligned} U - [(L \cap S \cap F) + (L \cap S) + (S \cap F) + (L \cap F) + (\text{only L players}) + (\text{only S players}) + (\text{only F players})] \\ = 255 - [11 + 80 + 25 + 56 + 13 + 17 + 20] \\ = 255 - 222 \\ = 33 \end{aligned}$$